

The cost of hedging XVA

Benedict Burnett and Ieuan Williams generalise the results of a 2021 article by the first author to obtain a formula for the hedging valuation adjustment (HVA) that is consistent with other XVAs. This HVA incorporates friction effects due to hedging both the underlying asset and counterparty credit, along with a contribution from hedge unwinds upon counterparty default. The numerical results for a simple case show that in a realistic situation the HVA can be on the same scale as more classical XVAs

In Burnett (2021), the concept of a hedging valuation adjustment (HVA) was introduced as a simple way to quantify the impact of friction costs on the value of a derivative book. The HVA introduced there was shown to resemble an XVA in the way that it acts as an adjustment to the risk-free value of the book. However, that article did not explore the interaction of this HVA with more standard XVAs. The aim of this article is to explore that interaction.

Although HVA may represent a relevant cost for any trading desk, it is perhaps the XVA desk of an institution for which it is likely to cause the most significant losses. This is for two main reasons, both related to the fact that hedging costs arise predominantly from second-order risks. First, a standard originating desk can hope to run a reasonably balanced book by pricing deals to encourage a reasonably flat net gamma position. Second, even if such a desk builds up significant gamma, this can often be efficiently semi-statically hedged by trading suitable options on the underlying asset.

For an XVA desk, the story is not so simple. First, this desk has much less opportunity to flatten its net gamma positions by encouraging particular trade types – if nothing else, this stems from the asymmetric form of many XVA adjustments (most XVAs act as a net cost to the institution). Second, due to the portfolio-level nature of XVAs, the desk will frequently find itself sitting on significant second-order cross risks that simply cannot be hedged with liquid options, because such options do not exist. For these and other reasons, the HVA can be a much bigger concern for an XVA desk than for many other desks in a bank. This is a fact known by XVA traders, but estimating the corresponding cost can be extremely challenging. In this article, we will see how the XVA desk's own HVA can be measured and handled in a consistent manner, and obtain some numerical results for a simple example case.

Setup

To obtain an XVA-consistent HVA, we must first define what we mean by XVA. These days, there exist a number of adjustments under this heading, including CVA (credit valuation adjustment), FVA (funding valuation adjustment), MVA (margin valuation adjustment (Green & Kenyon 2015)) and KVA (capital valuation adjustment (Green *et al* 2014)). In this paper, for the sake of having a reasonably simple framework that covers the primary points, we will limit ourselves to considering only CVA and FVA in the manner of Burgard & Kjaer (2013). The extension to other XVAs should be reasonably clear.

Deriving the HVA by considering a single counterparty on its own is not too hard; unfortunately, it is also unrealistic, since the XVA desk will hedge their net deltas aggregated across all counterparties as a single block, rather than hedge each counterparty separately. We must therefore consider the entire XVA book as a whole, ie, all counterparties at once, across their different originating books. We use 'book' as a catch-all term for any block of trades

that are hedged together by a trading desk. Each originating desk (eg, the interest rates trading desk) has its own book of trades covering multiple counterparties. The XVA desk's 'book' is defined to include all XVAs against all counterparties, and therefore includes the XVAs on all trades from all originating books. Thus, each counterparty will hit multiple 'books': its trades may sit with various originating desks, and will also be in the XVA book.

We will assume the HVA itself is hedged, along with its vanishing upon counterparty default.¹ For simplicity's sake, we will assume funding follows Strategy II from Burgard & Kjaer (2013), where a single own bond is used and the issuer's own default is not hedged. We will use the most naive possible extension of this strategy to multiple counterparties, to avoid distracting from the message; the coherence of the extension when treated properly is demonstrated in Burgard & Kjaer (2017). In line with these earlier treatments, we will assume deterministic rates, credit and funding spreads, again to keep the simplest possible formalism.

Given our previous knowledge, we expect each counterparty c to contribute a net (risk-free) mark-to-market (MtM) V_c and a net XVA U_c (key notation is summarised in table A). Along with these, we also expect each book j (including the XVA book) to yield a separate HVA H_j , since each book is taken to be hedged separately. Letting the HVA on the j th book be H_j , we can define the total value $\hat{V}$ of the (multi-counterparty, multi-book) derivative portfolio by:

$$\hat{V} = \sum_c (V_c + U_c) + \sum_j H_j \quad (\text{total value of the derivative portfolio})$$

$$H_j = \sum_c H_j^c \quad (\text{total HVA on book } j \text{ allocated to counterparty level})$$
(1)

For this derivatives portfolio, we will set up a hedge portfolio consisting of:

- χ units of the underlying asset S ;
- α_c units of each counterparty c 's zero-recovery bond P_c ;
- a cash account β_S secured against the asset, paying rate $(q_S - \gamma)$, where q_S is the repurchase agreement (repo) rate and γ is any dividend rate;
- a cash account β_c for each counterparty c , used to fund our position in that counterparty's bond and paying repo rate q_c ;
- a set of unsecured cash accounts β_c^F , used, respectively, to fund the MtM and (non-HVA) XVAs on counterparty c and all paying rate r_F ;
- a set of unsecured cash accounts β_j^F , each used, respectively, to fund the HVA on book j and paying rate r_F .

We neglect collateral; it should be clear how it could be added, but for now it would simply be a distraction. Specifically, its inclusion would affect the formulas for CVA and FVA (as previously shown in Burgard & Kjaer (2013)) but would not alter the final HVA formulas that are our target here.

¹ The extension to the case of an unhedged HVA was treated in Burnett (2021).

A. Summary of key notation used throughout the article

Symbol	Definition
$\hat{V}$	Total value of the derivative portfolio across all books and counterparties
V_c	Risk-free value of the derivatives traded with counterparty c
U_c	XVAs (excluding HVA) due to counterparty c
H_j	HVA due to book j
H_j^c	HVA due to book j allocated to counterparty c
S	Underlying asset (eg, stock)
$\sigma(S, t)$	Volatility of S
P_c	Zero-recovery bond issued by counterparty c
λ_c	Yield of counterparty c 's bond over repo rate q_c
r_F	The issuer's unsecured funding rate
J_c	Default indicator of counterparty c
$\Delta \hat{V}_c$	Change in $\hat{V}$ upon default of counterparty c
$d\phi$	Smoothed friction bleed in time dt
Ψ_c	Friction cost due to unwinding hedges upon default of counterparty c
m_j^c	Ratio of marginal HVA impact on book j upon default of counterparty c to its allocated value H_j^c
$f(t)$	Average rate of loss due to day-to-day friction costs
$\mathcal{D}_r(t, u)$	Rate r discount factor from time t to time u
$\psi_S(N)$	Friction cost of trading N units of the asset S
$\psi_c(N)$	Friction cost of trading N units of counterparty c 's credit risk
Γ_j, γ_j^c	Total asset S gamma on book j , and the amount due to counterparty c
$\Gamma_{SS}, \gamma_{SS}^c$	Total asset S gamma on the XVA book, and the amount due to counterparty c
Γ_{Sc}	Asset versus credit cross-gamma on the XVA book due to counterparty c
δ_c	The asset delta on counterparty c 's mark-to-market, $\partial_S V_c$

The total portfolio value is then:

$$\Pi = \chi S + \sum_c \alpha_c P_c + \beta_S + \sum_c \beta_c + \sum_c \beta_c^F + \sum_j \tilde{\beta}_j^F + \hat{V} \quad (2)$$

The different holding amounts here are related by:

$$\beta_S = -\chi S, \quad \beta_c = -\alpha_c P_c, \quad \beta_c^F = -(V_c + U_c), \quad \tilde{\beta}_j^F = -H_j \quad (3)$$

while the various dynamics are given by the following set of equations:

$$\begin{aligned} dS &= \mu(S, t) dt + \sigma(S, t) dW_t, & dP_c &= (q_c + \lambda_c) P_c dt - P_c dJ_c \\ d\beta_S &= (q_S - \gamma) \beta_S dt, & d\beta_c &= q_c \beta_c dt \\ d\beta_c^F &= r_F \beta_c^F dt, & d\tilde{\beta}_j^F &= r_F \tilde{\beta}_j^F dt \end{aligned}$$

Here, each dJ_c is a Poisson jump process. For simplicity in the derivation, we take all counterparty default processes to be independent: both of one another and of the market dW_t . We neglect the own-default risk of the funding accounts, since (following Burgard & Kjaer) we will not consider the hedging of our own default (although note that the issuer's own credit risk will materially affect its funding rate r_F).

A change of view

In what follows, it will be helpful to use an alternative view of the formalism in Burnett (2021). The results there were derived by considering a single reheding interval δt at a time. However, the same HVA expressions could have been attained by considering an infinitesimal time step dt , with the friction costs smoothed into a continuous bleed at the 'friction rate' f . The formalism under such an approach is artificial, since one ends up pretending that the portfolio is perfectly hedged at any instant; in fact, this is, of

course, only true at the moments of reheding. However, the infinitesimal view works because there are fundamentally two effects due to real-world hedging: (1) P&L variance due to not being perfectly hedged the whole time and (2) a net P&L shift due to the cost of friction itself. The HVA is defined by effect (2) alone; the effects of not being perfectly hedged at all times average away in its derivation. We could thus perform the same derivation as in Burnett (2021), essentially replacing equation (4) there with:

$$\mathbb{E}_t[d\Pi] = (\hat{V}_t + r \chi \hat{V}_x + \frac{1}{2} \sigma^2 \hat{V}_{xx} - r \hat{V}) dt - \mathbb{E}_t[d\phi] = 0 \quad (4)$$

with $d\phi$ being the infinitesimal 'smoothed' friction bleed in time dt under this transformation. The result of such an approach is identical to the discrete method applied in that paper, despite the fact that the portfolio will not in reality be self-financing.

This infinitesimal view will be useful in what follows, to allow us to deal with the multiple possible reheding intervals that come into play.

Hedging strategy

With the above in mind, the change in our XVA-aware portfolio value (2) over time dt is given by:

$$\begin{aligned} d\Pi &= \left(\hat{V}_t + \frac{1}{2} \sigma^2 \hat{V}_{SS} + \sum_c \alpha_c P_c \lambda_c - (q_S - \gamma) \chi S - r_F \hat{V} \right) dt \\ &\quad + (\chi + \hat{V}_S) dS + \sum_c (\Delta \hat{V}_c - \alpha_c P_c) dJ_c - d\phi \end{aligned} \quad (5)$$

where $\Delta \hat{V}_c$ represents the change in derivative portfolio value upon default of counterparty c , and $d\phi$ is the friction bleed in time dt as above.

Now we want to hedge our S and J_c risks as far as possible. The first of these can be hedged as usual by setting $\chi = -\hat{V}_S$. However, for the second, observe that the default of any counterparty c will itself incur a friction cost due to the necessity of unwinding our hedges, which we should not neglect here; thus, $d\phi$ will contain a dJ_c component. Letting ϕ_{drag} represent the remaining component of friction costs due to standard day-to-day hedging (ie, due to market moves rather than defaults), the total friction term in (5) can therefore be decomposed as:

$$d\phi = d\phi_{\text{drag}} + \sum_c \Psi_c dJ_c \quad (6)$$

Given this, we see that counterparty defaults can be hedged by setting $\alpha_c P_c = \Delta \hat{V}_c - \Psi_c$.

As explained above, although these hedge formulas properly apply only at the instant of hedging each risk, we will neglect this subtlety here, as it comes to the same effect in expectation as our current approach (and expectation is all we will be interested in). The hedged change in value therefore becomes:

$$\begin{aligned} d\Pi &= \left(\hat{V}_t + (q_S - \gamma) S \hat{V}_S + \frac{1}{2} \sigma^2 \hat{V}_{SS} - r_F \hat{V} + \sum_c \lambda_c (\Delta \hat{V}_c - \Psi_c) \right) dt \\ &\quad - d\phi_{\text{drag}} \end{aligned} \quad (7)$$

Closeout

To find $\Delta \hat{V}_c$, we consider (a) what must be subtracted from the total value $\hat{V}$ in the instant of c 's default, along with (b) what we stand to make back just afterwards due to closeout. The latter we take to be the standard closeout value:

$$g_c = R_c V_c^+ - V_c^- \equiv V_c - (1 - R_c) V_c^+ \quad (8)$$

with R_c the recovery rate of counterparty c . (This assumes XVAs are all lost at closeout. We leave to one side the interesting related questions here.)

To see what gets removed at the instant of default, clearly we must at least remove the originating desk value V_c , along with all the standard XVAs U_c . But the HVA removal is harder, since the total HVA is not an independent sum over counterparties (as demonstrated in Burnett (2021)). Instead, we must consider the HVA impact of removing counterparty c 's trades; this is equivalent to the marginal impact of that counterparty on each HVA H_j , which will in general not be identical to the allocated HVA H_j^c as defined in (1). For now, we will therefore denote the marginal impact by some multiplier m_j^c times the allocated HVA for each counterparty c and book j . (For the vanilla case, Burnett (2021) showed that this multiplier was equal to 2; below, we will see that the full picture can be somewhat more complex.)

We therefore have:

$$\Delta \hat{V}_c = -(1 - R_c)V_c^+ - U_c - \sum_j m_j^c H_j^c \quad (9)$$

Given this, (7) becomes:

$$d\Pi = \left[\hat{V}_t + (q_S - \gamma)S\hat{V}_S + \frac{1}{2}\sigma^2\hat{V}_{SS} - r_F\hat{V} - \sum_c \lambda_c \left((1 - R_c)V_c^+ + U_c + \sum_j m_j^c H_j^c + \Psi_c \right) \right] dt - d\phi_{\text{drag}} \quad (10)$$

As in Burnett (2021), we define the HVA as being whatever is needed to make the expected P&L over each time step equal to zero (in the real-world measure). This gives us:

$$0 = \left[\hat{V}_t + (q_S - \gamma)S\hat{V}_S + \frac{1}{2}\sigma^2\hat{V}_{SS} - r_F\hat{V} - \sum_c \lambda_c \left((1 - R_c)V_c^+ + U_c + \sum_j m_j^c H_j^c + \Psi_c \right) \right] dt - \mathbb{E}_t[d\phi_{\text{drag}}] \quad (11)$$

Following Burnett (2021) and extending into the infinitesimal case, we can now define the 'friction rate' as the average rate of loss due to day-to-day friction:

$$f(t) = \frac{\mathbb{E}_t[d\phi_{\text{drag}}]}{dt} \quad (12)$$

We take the total originating desk value to be defined by:

$$V_t + (q_S - \gamma)SV_S + \frac{1}{2}\sigma^2V_{SS} - r_oV = 0 \quad (13)$$

where r_o is the rate used by the originating desks to discount their valuations. Subtracting this from (11) and rearranging gives:

$$\begin{aligned} & \sum_c (\partial_t + \mathcal{A}_t - (r_F + \lambda_c))U_c + \sum_{j,c} (\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c))H_j^c \\ &= (r_F - r_o)V + \sum_c \lambda_c(1 - R_c)V_c^+ + \sum_c \lambda_c \Psi_c + f \quad (14) \end{aligned}$$

with $\mathcal{A}_t$ the generator of the risk-neutral process.

Distilling to HVA

At this point, we can separate out the classic XVAs U_c . Defining:

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + \lambda_c))U_c = (r_F - r_o)V + \sum_c \lambda_c(1 - R_c)V_c^+ \quad (15)$$

we obtain the Burgard and Kjaer FVA and CVA for each counterparty:

$$\begin{aligned} U_c &= - \int_t^T \mathbb{E}_t^{\mathbb{Q}}[\mathcal{D}_{r_F + \lambda_c}(t, u)(r_F - r_o)V_c] du \\ &\quad - \int_t^T \mathbb{E}_t^{\mathbb{Q}}[\mathcal{D}_{r_F + \lambda_c}(t, u)\lambda_c(1 - R_c)V_c^+] du \quad (16) \end{aligned}$$

where:

$$\mathcal{D}_r(t, u) \equiv \exp\left(-\int_t^u r(s) ds\right)$$

and the expectations are in the risk-neutral measure $\mathbb{Q}$.

This being done, we can subtract (15) from (14), leaving us with an equation for the combined HVAs:

$$\sum_{j,c} (\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c))H_j^c = f + \sum_c \lambda_c \Psi_c \quad (17)$$

At this point, we must note that the day-to-day 'drag' friction bleed f is made up of two different types of component: an 'asset friction' f_j^S , due to hedging the S -dependence of book j (ie, friction losses from buying and selling the asset S), and a 'credit friction' f_c^λ , from hedging the credit of counterparty c (ie, friction losses from buying and selling the bond P_c , or equivalent instruments). A more explicit version of the previous equation is therefore:

$$\sum_{j,c} (\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c))H_j^c = \sum_j f_j^S + \sum_c f_c^\lambda + \sum_c \lambda_c \Psi_c \quad (18)$$

It will be clearest in what follows to distinguish the XVA desk from the originating desks, as their HVAs will be qualitatively different. To this end, we can denote the XVA desk's quantities with an index X , leaving the general index j to refer to the range of originating desks. The previous equation thus expands to become:

$$\begin{aligned} & \sum_{c, j \neq X} (\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c))H_j^c \\ &+ \sum_c (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c))H_X^c \\ &= \sum_{j \neq X} f_j^S + f_X^S + \sum_c f_c^\lambda + \sum_c \lambda_c \Psi_c \quad (19) \end{aligned}$$

■ **Originating desk HVAs.** We can now begin to titrate the different HVAs. We start by separating the HVA carried by the XVA desk (let this be H_X^c) from those carried by originating desks. Here, we will assume all counterparty credit and closeout costs are borne by the XVA desk, meaning the related costs f_c^λ and Ψ_c will be paid only by that desk.

The HVA of each originating desk j due to counterparty c is therefore given by:

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + m_j^c \lambda_c))H_j^c = f_j^S \quad (20)$$

A formula for the friction rate here was derived in Burnett (2021) for the case where the desk's hedging strategy is based on a delta threshold, ie, when

the net delta on desk j 's position exceeds a threshold $\pm D_j$, they flatten their book. Maintaining that model here, we have:

$$f_j^S = \left(\frac{\sigma \Gamma_j}{D_j} \right)^2 \psi_s(D_j) \quad (21)$$

where Γ_j is the book-level gamma and $\psi_s(N)$ represents the friction cost of trading N units of the asset S (which may be, for example, a constant spread times the quantity transacted). As in Burnett (2021), we will for simplicity assume that the HVA itself has a negligible effect on each book's gamma, so as to allow a closed-form solution. Following the same gamma-allocation approach as in that paper, this gives us the (XVA-consistent) originating desk HVA per counterparty below:

$$H_j^c = - \int_t^T \mathbb{E}_t^Q \left[\mathcal{D}_{r_F + m_j^c \lambda_c}(t, u) \sigma^2 \frac{\psi_s(D_j)}{D_j^2} \Gamma_j \gamma_j^c \right] du \quad (22)$$

where we acknowledge a separate gamma contribution γ_j^c from each counterparty. In other words, the HVA on each counterparty is proportional to the product of the book and counterparty-level gammas. This is very similar in shape to the original HVA from Burnett (2021), the only visible difference being the discount rate. This similarity indicates that the correct value of the m_j^c factor is approximately 2, driven by the quadratic dependence of the book HVA on the net gamma. The resulting 'super-contingency' of the adjustment can be seen as stemming from the fact that in the case of c 's default, not only does their HVA H_j^c vanish, but all other counterparties' HVAs are also affected through the impact on the book-level Γ_j .

The correct treatment of the Γ_j factor here depends somewhat on choices made in the institution's setup. We do not have space to explore such points here, but note that the book-level Γ_j should in general be default-aware (ie, a sum of the form $\sum_c (1 - J_c) \gamma_j^c$, reducing to $\sum_c \mathcal{D}_{\lambda_c}(t, u) \gamma_j^c$ within the expectation).

■ **XVA desk HVA.** Having found the originating desks' HVAs, we are left with the part H_X^c that falls to the XVA desk. Subtracting (20) from (19), we are left with:

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_X^c = f_X^S + \sum_c f_c^\lambda + \sum_c \lambda_c \Psi_c \quad (23)$$

The contributions to the right-hand side are of two distinct kinds. In the first two terms, we have the costs from standard day-to-day hedging of the underlying asset and credit, which are easily understood using the work of Burnett (2021). The third term is qualitatively different, coming from counterparty closeouts. Let us therefore temporarily separate the XVA desk's HVA into these two aspects: a 'drag HVA' H_{Drag}^c , covering the day-to-day friction costs, and a 'closeout HVA' H_{Closeout}^c , due to the costs upon counterparty closeout:

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Drag}}^c = f_X^S + \sum_c f_c^\lambda \quad (24)$$

$$\sum_c (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Closeout}}^c = \sum_c \lambda_c \Psi_c \quad (25)$$

■ **HVA aspect I: drag HVA.** The drag component of HVA can be handled very much as before: it is the multi-asset equivalent of (22), this time including all the counterparty hazard rates as assets as well as the underlying S . At this point, we will drop the X indexing, as it is clear we are only talking about the quantities handled by the XVA desk.

We can expand the friction rates from (24), taking advantage of the work done in Burnett (2021) on the multi-asset case to see that the credit friction stems entirely from the asset versus credit cross-gamma Γ_{sc} and movements in the underlying asset (due to the lack of a Brownian factor in the bond price). Note that we are here assuming the same delta-threshold-per-asset hedging strategy studied in that paper; as stated there, the expression for friction rates will be sensitive to the assumed hedging strategy.

Let s subscripts denote quantities connected to the underlying asset and c subscripts denote the analogous quantities related to the credit of counterparty c (so, for example, D_c is the delta threshold for hedging credit delta). We let Γ_{ss} represent the book-level asset gamma on the XVA and γ_{ss}^c represent that part due to counterparty c . We therefore have:

$$\begin{aligned} \sum_c (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Drag}}^c &= \left(\frac{\sigma \Gamma_{ss}}{D_s} \right)^2 \psi_s(D_s) + \sum_c \left(\frac{\sigma \Gamma_{sc}}{D_c} \right)^2 \psi_c(D_c) \\ &= \sum_c \left(\frac{\psi_s}{D_s^2} \sigma^2 \Gamma_{ss} \gamma_{ss}^c + \frac{\psi_c}{D_c^2} \sigma^2 \Gamma_{sc}^2 \right) \end{aligned} \quad (26)$$

Note that the nature of the assumed setup, with counterparties being treated independently, means that the asset versus credit cross-gamma Γ_{sc} is already a per-counterparty quantity that does not require further allocation.² We will maintain the simplifying assumption that the HVA itself has a negligible impact on the book-level gamma.

We therefore obtain:

$$\begin{aligned} H_{\text{Drag}}^c &= - \int_t^T \mathbb{E}_t^Q \left[\mathcal{D}_{r_F + m_X^c \lambda_c}(t, u) \right. \\ &\quad \left. \times \left(\frac{\psi_s}{D_s^2} \sigma^2 \Gamma_{ss} \gamma_{ss}^c + \frac{\psi_c}{D_c^2} \sigma^2 \Gamma_{sc}^2 \right) \right] du \end{aligned} \quad (27)$$

Similar remarks apply here to those above regarding the proper form of the Γ terms: depending on the institution's setup, these should properly incorporate the effect of defaults in the book, although such features may be beyond the reach of a simple treatment.

We note that (27) is effectively just an application of the allocation formula from Burnett (2021). The general multi-asset drag HVA would therefore be:

$$\begin{aligned} H_{\text{Drag}}^c &= - \sum_i \int \mathbb{E} \left[\mathcal{D}_{r_F + m_X^c \lambda_c}(t, u) \frac{\psi_i(D_i)}{D_i^2} \sum_{j,k} \rho_{jk} \sigma_j \sigma_k \Gamma_{ij} \gamma_{ik}^c \right] du \\ &\equiv - \int \mathbb{E} [\mathcal{D}_{r_F + m_X^c \lambda_c}(t, u) \text{tr}(\mathbf{F} \mathbf{\Gamma} \mathbf{\Sigma} \mathbf{\gamma}^c)] du \end{aligned} \quad (28)$$

where $\mathbf{F} = \text{diag}(\psi_1/D_1^2, \dots, \psi_N/D_N^2)$, $\mathbf{\Gamma}$ is the XVA book-level Hessian, $\mathbf{\gamma}^c$ is that of counterparty c , and $\mathbf{\Sigma}$ is the covariance matrix.

■ **HVA aspect II: closeout HVA.** Finally, we are left with the closeout component of HVA coming from the $\lambda_c \Psi_c$ term in (25), where Ψ_c represents the friction cost of unwinding a component of our hedges upon the default of counterparty c .

If c defaults, we will need to unwind our asset (S) hedge against them, but not the bond hedge, which will be used directly to offset our losses. The amount of asset hedge to unwind is given by the marginal contribution of counterparty c to the hedge holding, which (following very similar reasoning

² This would be complicated by index-based credit hedging, which we neglect here.

to before) is given by:

$$\chi_c = -\partial_S \left(V_c + U_c + \sum_j m_j^c H_j^c \right) \quad (29)$$

The friction cost of unwinding this is then $\Psi_c = \psi_s(|\chi_c|)$. Substituting this into (25), we have:

$$\begin{aligned} (\partial_t + \mathcal{A}_t - (r_F + m_X^c \lambda_c)) H_{\text{Closeout}}^c \\ = \lambda_c \psi_s \left(\partial_S V_c + \partial_S U_c + \sum_j m_j^c \partial_S H_j^c \right) \end{aligned} \quad (30)$$

As before, we can obtain a closed-form solution here by making the simplifying assumption that the risk on the HVA itself is negligible compared with the total delta. This allows us to neglect the $\partial_S H_j^c$ contribution on the right-hand side. A reasonable approximation is also likely to be given by neglecting the $\partial_S U_c$ contribution, since it is ‘higher order’ than the underlying $\partial_S V_c$ contribution. Given the whole is multiplied by a hazard rate, this approximation is on the same level as neglecting XVA contributions to closeout when calculating the CVA. For shorthand, let $\delta_c \equiv \partial_S V_c$, the delta on counterparty c ’s MtM. We can then obtain the closeout HVA on each counterparty as:

$$H_{\text{Closeout}}^c \approx - \int_t^T \mathbb{E}_t^Q [\mathcal{D}_{r_F + m_X^c \lambda_c}(t, u) \lambda_c \psi_s(|\delta_c|)] du \quad (31)$$

The total HVA seen by the XVA desk is finally the sum of drag and closeout contributions; we can thus combine (27) and (31) to obtain:

$$\begin{aligned} H_X^c = - \int_t^T \mathbb{E}_t^Q \left[\mathcal{D}_{r_F + m_X^c \lambda_c}(t, u) \right. \\ \left. \times \left(\frac{\psi_s(D_s)}{D_s^2} \sigma^2 \Gamma_{ss} \gamma_{ss}^c + \frac{\psi_c(D_c)}{D_c^2} \sigma^2 \Gamma_{sc}^2 + \lambda_c \psi_s(|\delta_c|) \right) \right] du \end{aligned} \quad (32)$$

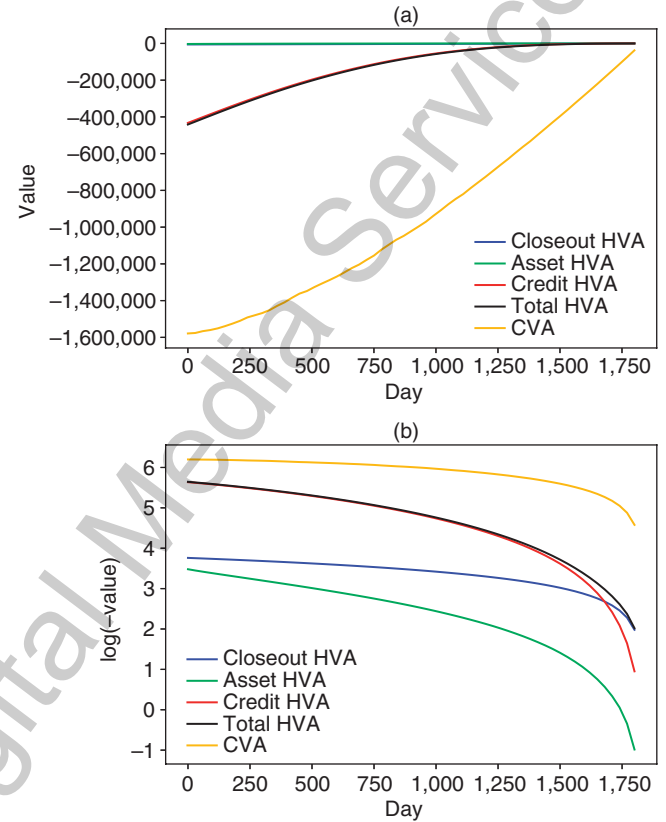
This XVA-consistent HVA can then be added to the other XVAs U_c to form part of the total value of a portfolio, following (1).

Note that the correct value of the XVA desk’s m_X^c factor depends on the relative size of the drag and closeout components of HVA for counterparty c , but it should generally sit between 1 and 2. Under the assumption that the drag effect is dominant, a value of 2 will generally not be far off the correct setting.

Numerical results

Here, we present the results of a simple implementation of HVA as an XVA. We use a Black-Scholes formulation to price the CVA of a book containing a single counterparty with a single five-year, 1 billion foreign notional FX forward; this is employed within a Monte Carlo simulation to calculate (32). For the FX, we use a spot of 0.7, a lognormal volatility of 7.5%, a delta threshold D_s of 50,000 domestic currency units per percent and a friction cost (spread to mid) of 1 pip. For credit, we take the credit spread to be fixed at 100 basis points with 40% recovery, a delta threshold D_c of 5,000 domestic currency units per bp, and a friction cost of 5bps (note that credit friction will in reality be highly dependent on the liquidity of the specific instrument being traded). We set all interest rates to zero for simplicity, thus removing FVA from consideration. In line with the above formalism, we assume no collateral (a more thorough study of different portfolios, collateral arrangements

1 The run-down profile of each component of the HVA over time, plotted along with the CVA itself, for the case described in ‘Numerical results’



It can be seen that the credit component of HVA dominates over the asset and closeout components (the credit component is almost hidden by the total HVA line). The second plot shows the same in log space, with logarithms taken to base 10

and market conditions would warrant a dedicated treatment in future). The linearity of an FX forward renders the originating desk’s HVA negligible; we thus focus on the HVA of the XVA desk alone.

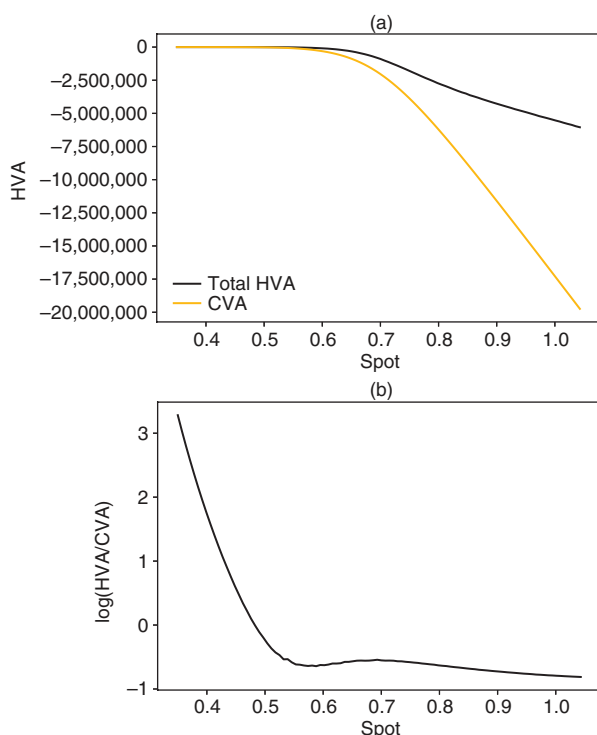
It will be helpful in what follows to separate out the costs of hedging the underlying asset and those of hedging credit. To that end, we dub the first contribution in (32) (dependent on ψ_s) the ‘asset HVA’ and the second contribution (dependent on ψ_c) the ‘credit HVA’.

Figure 1 shows the resulting HVA components alongside the CVA. Starkly visible in this figure is the fact that the total HVA (in black) is dominated by the credit component (as one might expect, given the difference in liquidity between credit and FX markets), so much so that its other components are barely distinguishable.³ We see that, for this five-year deal, the initial HVA represents almost a 30% surcharge in addition to CVA. This gives a hint at the potential scale of such an adjustment on a larger XVA portfolio, which would likely have substantially larger gammas, a longer duration and sensitivities to many more market instruments.

It is also of some interest to consider the size of the HVA sensitivities in comparison to the XVA risks.

³ This may not always be the case when considering the marginal impact of a single trade on a large book, where we could expect the ‘asset HVA’ component to increase significantly due to the presence of the book gamma in (32).

2 Variation of HVA and CVA with spot level (ie, moneyness) for the setup described in 'Numerical results'



We show both absolute values and the log of the ratio, with log taken to base 10

Figure 2 shows how HVA behaves as a function of the moneyness of the CVA for our example case. It can be seen that for a very small (out-of-the-money) CVA, the HVA can significantly dominate over the CVA, raising important questions about the best hedging strategy in such a situation (although the absolute values of both adjustments are relatively small when this occurs).

Conclusions

In this article, we have developed and extended the treatment of HVA introduced in Burnett (2021), showing how it can be handled in a consistent framework with standard XVAs and multiple books. We have seen that HVA brings some peculiarities not seen in other XVAs, such as its closeout aspect (due to the cost of unwinding hedges upon counterparty default), but it can be treated in a similar manner overall. We have also seen that the impact of HVA can be comparable to that of CVA in a simple example case, suggesting that we cannot realistically afford to neglect it from consideration.

An important point to note is that the bulk of real-world HVA is likely to come from cross-gammas rather than from pure gamma risks. We have seen this in the numerical results for the case of credit cross-gamma, but the point applies more widely: the key reason is that a pure gamma can often be hedged with a suitable option, virtually annihilating its contribution to future hedge costs. However, cross-gammas generally represent risks that are far harder to hedge with options (due to the lack of suitable liquid hybrids), meaning that a trader cannot do much better than follow the kind of delta-hedging strategy considered here. This is a problem that particularly afflicts the XVA space due to its cross-asset nature.

It is worth taking a step back from the formalism and considering the value of HVA from a trader's point of view. Upon inception of a deal, a trader in general needs to consider whether the new trade makes sense for the institution at a given price. The original introduction of adjustments such as CVA broadened this question to include aspects of credit risk: a new deal might look very attractive, but if there is a large risk of the client defaulting during the life of the trade, its appeal – and hence its value – may be dramatically reduced. Similarly, FVA and KVA brought to the fore the questions of the lifetime funding and capital costs implicit in a new deal. In each case, such costs were by no means unknown to traders beforehand; the relevant adjustments simply served to crystallise the questions and provide a precise measurement of the impact of each consideration on financial value.

In the case of hedging, we have a potentially similar picture. Traders know friction will cost them money over the life of any new trade, and they know this cost will be driven by the gamma and cross-gamma impacts of the trade. The difficulty, from that knowledge, is in determining how some particular combination of cross-gamma impacts will combine to affect total hedging costs. The XVA impact of a new trade could easily carry a dozen second-order risks; knowing how to project these onto the single axis of 'cost' is in general extremely challenging. It is our hope that the HVA formalism developed here provides a first step in solving exactly that problem, which is already of daily relevance to traders.

As implied, this is clearly not the end of the story. Much work remains to be done on considering the impact of different hedging strategies, and particularly on determining the expected variance of P&L under each given strategy. There are also avenues to explore regarding the incorporation of other XVAs, and concerning the efficient estimation of HVAs, due to the computational challenges of integrating over simulated XVA gammas in (32), for example. Addressing these questions will lend precision to, but will not change, the basic picture: HVA measures a very real cost faced by financial institutions due to their fundamental business of mitigating risk. ■

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